Sahaj Bhadja

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Summary

Application engineer with 2+ years building **platform-independent GUI software and modular architectures** using TypeScript, JavaScript, React, Angular.js, Node.js, Git, Agile development, continuous testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Designed modular FastAPI applications with separated orchestration, storage, and validation concerns, processing 5,000+ monthly jobs while improving maintainability, failure isolation, and production reliability to 99%+ at scale.
- Analyzed and debugged application performance for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through documented query optimization, indexing, testing, and iterative architecture improvements for reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed responsive React GUI features for file upload, grading history, and question management, integrating Flask REST APIs while serving 150+ instructors and 10,000+ daily requests reliably at scale.
- Implemented a reusable switchable architecture across OpenAI, Anthropic, and Qwen providers, adding concurrent processing, automated testing, and fallback behavior while reducing user feedback turnaround 60% for instructors.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed modular Python software and evaluation interfaces processing 100,000+ trajectory logs, improving long-horizon planning 18% through separation of data processing, model components, simulation, and reproducible performance testing.
- Collaborated on system requirements, organized experiment tasks, and documented architecture evaluations using Hydra and Weights & Biases, improving traceability and closed-loop software stability 15% during simulation at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Designed and tested platform-independent microservice software across Windows, macOS, and Linux, automating 20+ unit and integration tests with Jenkins, PyTest, and Mockito while increasing pipeline efficiency 68%.
- Collaborated with Microsoft and GitHub stakeholders in Agile sprints to define requirements, debug Java and Python integrations, and deliver maintainable solutions that improved API response times 30%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed React and TypeScript application interfaces backed by Node.js, WebSockets, and REST APIs, synchronizing two million records with 33% fewer conflicts through eight highly cohesive modular services.
- Participated in code review, testing, and staged releases using Git, Jest, Selenium, Docker, and Jenkins, reliably reducing review turnaround 25% and deployment time 50% across application environments.

Skills

GUI Development: TypeScript, JavaScript, React, Angular.js, Next.js, HTML, CSS, responsive interfaces
Application Systems: Node.js, Express.js, REST APIs, WebSockets, Java, Python, FastAPI, Flask
Architecture: Separation of concerns, modular design, high cohesion, low coupling, OOP, microservices
Quality: Unit testing, integration testing, continuous testing, Jest, PyTest, Mockito, Selenium
Engineering: Git, Agile/Scrum, CI/CD, Jenkins, Linux, Windows, macOS, debugging, documentation

Projects

Chat2SQL Code | *TypeScript, Cross-Platform GUI*

- Built a platform-independent Java and TypeScript database GUI supporting 10+ systems, implementing schema synchronization and Docker-based deployment across Windows, macOS, and Linux with reusable frontend and backend modules.

PiXMod Live | *Angular.js, Next.js, UI Development*

- Developed a responsive Angular.js and Next.js application interface for AI image workflows, integrating CLIP, SAM, Stable Diffusion, MongoDB storage, semantic search, and Stripe subscriptions across modular features.

CPU-scheduling Simulator Live | *GUI Software, OS Fundamentals*

- Implemented six operating-system scheduling algorithms with responsive React controls, automated waiting and turnaround metrics, and dynamic Gantt visualization, testing 1,000+ process combinations across preemptive and non-preemptive scenarios.